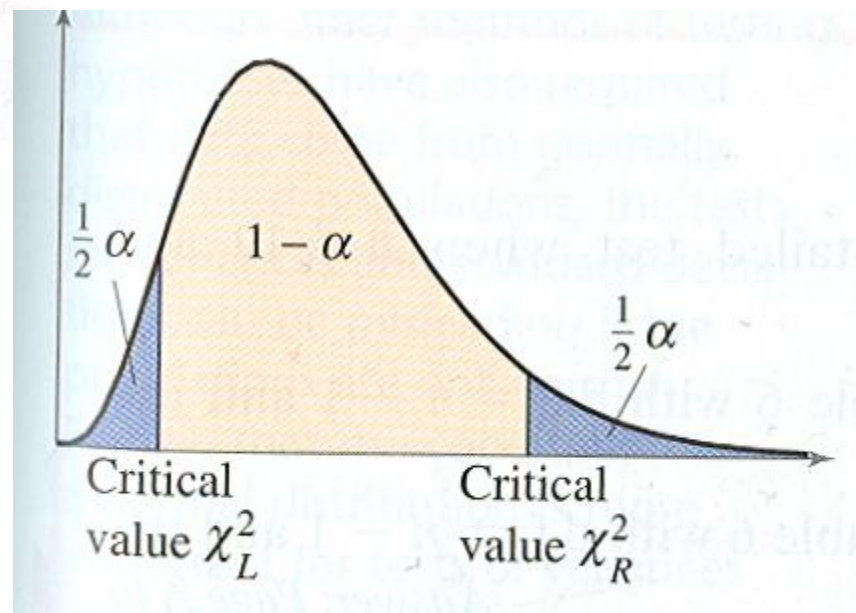
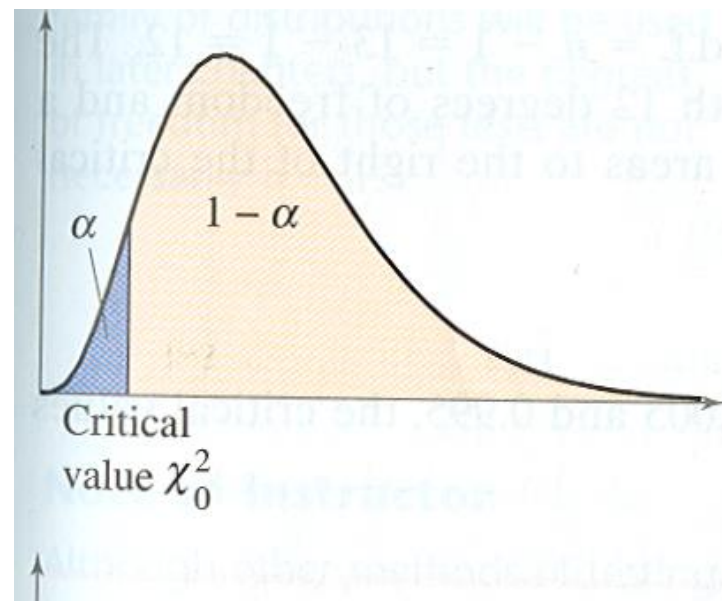
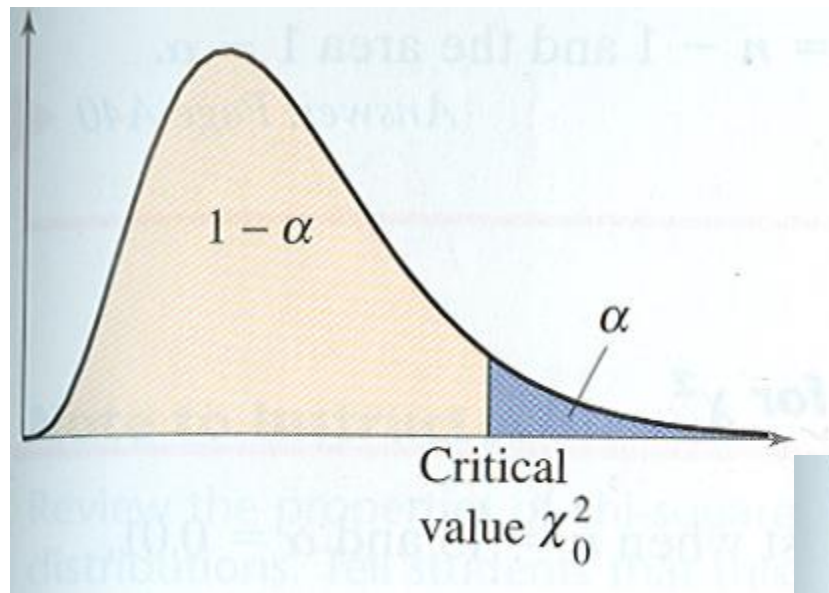


Hypothesis Testing for the Variance and Standard Deviation

Critical Values

- In “REAL” life, it is often important to produce consistent predictable results. For instance, consider a company that manufactures golf balls. The manufacturer must produce millions of golf balls each having the same size and the same weight. There is a very low tolerance for variation. If the population is normal, you can test the variance and standard deviation of the process using the chi-square distribution with $n - 1$ degrees of freedom.

Reminder:



GUIDELINES

Finding Critical Values for the χ^2 -Test

1. Specify the level of significance, α .
2. Determine the degrees of freedom, $\text{d.f.} = n - 1$.
3. The critical values for the χ^2 -distribution
To find the critical value(s) for a
 - a. *right-tailed test*, use the value that corresponds to d.f. and α .
 - b. *left-tailed test*, use the value that corresponds to d.f. and $1 - \alpha$.
 - c. *two-tailed test*, use the values that correspond to d.f. and $\frac{1}{2}\alpha$ and d.f. and $1 - \frac{1}{2}\alpha$.

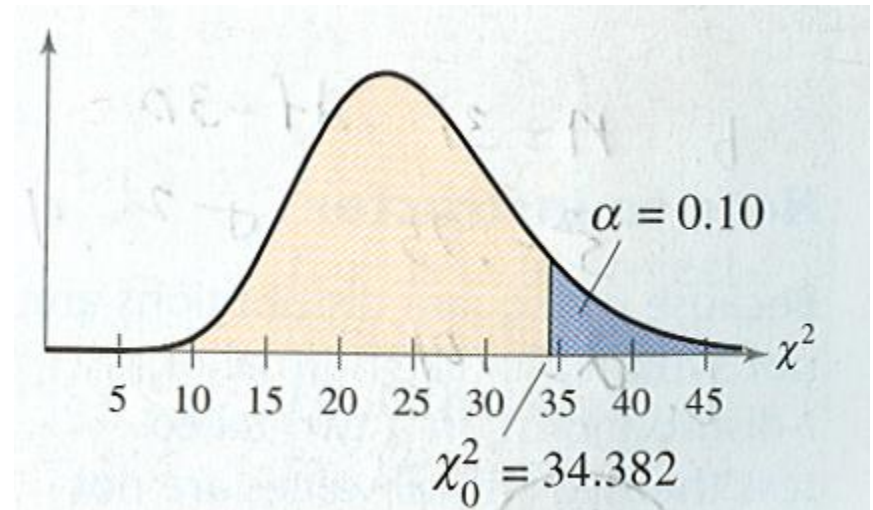
Ex. 1: Finding Critical Values for χ^2

- Find the critical χ^2 value for a right-tailed test when $n = 26$ and $\alpha = 0.10$
- SOLUTION: The degrees of freedom are d.f. = $n - 1 = 26 - 1$. The graph on the next slide shows an 25 degrees of freedom and a shaded area of $\alpha = 0.10$ in the right tail.

Ex. 1: Finding Critical Values for χ^2

- Using Table 6 with d.f. = 25 and $\alpha = 0.10$, the critical value is:

$$\chi^2_0 = 34.382$$



Look – Chi Table

locate DF, and $\alpha = 0.10$.
Where they meet up is
the value.

Ex. 2: Finding Critical Values for χ^2

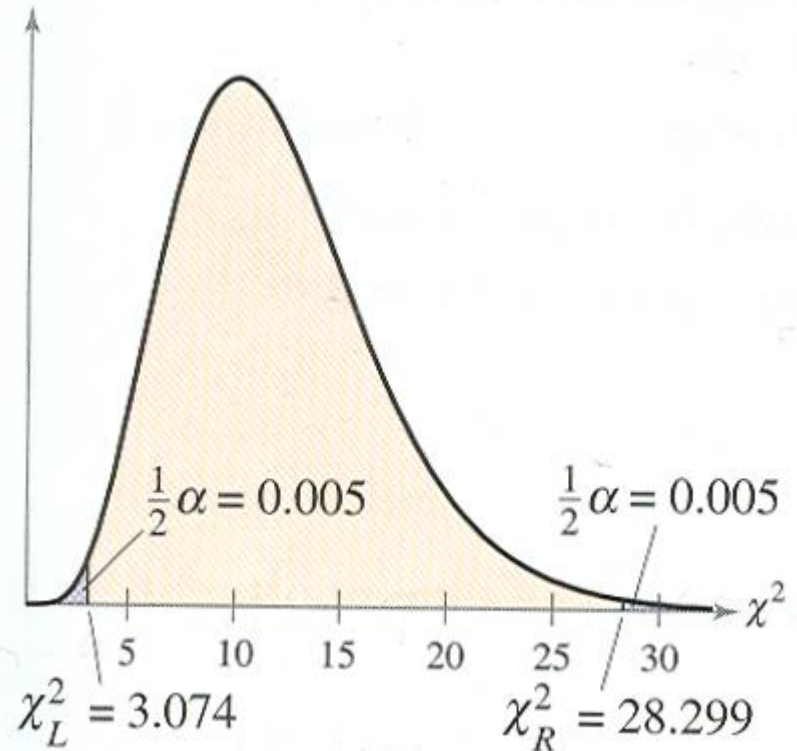
- Find the critical value χ^2 – value for a left-tailed test when $n = 11$ and $\alpha = 0.01$.

Solution: The degrees of freedom = $n - 1 = 11 - 1 = 10$.
The graph on the next slide shows a χ^2 – distribution with 10 degree of freedom and a shaded area of $\alpha = 0.01$ in the left tail. The area to the right of the critical value is $1 - \alpha$; so $1 - 0.01 = 0.99$.

Ex. 2: Finding Critical Values for χ^2

- Using Table 6 with d.f. = 10 and area = is: $1 - 0.01 = 0.99$, the critical value is:

$$\chi^2_0 = 2.558$$



Look – Chi Table

locate DF=10, and $\alpha = 0.01$. Where they meet up is the value.

Note to yourself

- Because chi-square distributions are not symmetric (like normal distribution or t-distributions, in a two-tailed test, the two critical values are NOT opposites. Each critical value must be calculated separately.

Finding Critical Values for X^2

- Find the critical value X^2 – value for a two-tailed test when $n = 13$ and $\alpha = 0.01$.

Solution: The degrees of freedom = $n - 1 = 13 - 1 = 12$.
The graph on the next slide shows a X^2 – distribution with 12 degree of freedom and a shaded area of $\alpha = 0.005$ in the left tail. The area to the right of the critical value is $\frac{1}{2}\alpha = 0.005$

- In each tail. The areas to the right of the critical values are

$$\frac{1}{2}\alpha = 0.005$$

$$1 - \frac{1}{2}\alpha = 0.995$$

Ex. 3: Finding Critical Values for χ^2

- Using Table 6, with d.f. = $n - 1 = 13 - 1 = 12$ and the areas 0.005 and 0.995, the critical values are:

$$\chi^2_L = 3.074 \qquad \chi^2_R = 28.299$$

Chi-Square Test

- To test a variance σ^2 or a standard deviation σ of a population, that is normally distributed, you can use the χ^2 -test

χ^2 -Test for a Variance or Standard Deviation

The χ^2 -test is a statistical test for a population variance or standard deviation. The χ^2 -test can be used when the population is normal. The **test statistic** is s^2 and the **standardized test statistic** is χ^2 . The sampling distribution for s^2 is a chi-square distribution with $n - 1$ degrees of freedom.

$$\chi^2 = \frac{(n - 1)s^2}{\sigma^2}$$

Step 1: State the claim mathematically and verbally. Identify the null and alternative hypotheses.

Ha and Ho

Step 2: Find the test statistic. $\chi^2 = \frac{(n-1)s^2}{\sigma^2}$

step 3: find the critical value from the table

step 4: Make a decision to accept or reject

Ha

Reminder:

- The tests for variance and standard deviation can be misleading if the populations are NOT normal. It is the condition for a normal distribution is more important for tests of variance or standard deviation.

Ex. 1: Using a Hypothesis Test for the Population Variance

- A dairy processing company claims that the variance of the amount of fat in the whole milk processed by the company is no more than 0.25. You suspect that is wrong (**our claim**) and find a random sample of 41 milk containers has a variance of 0.27. At $\alpha = 0.05$, is there enough evidence to reject the company's claim? **Assume the population is normally distributed.**

Solution:

- The claim is “the variance is no more than 0.25.” So, the null and alternative hypotheses are:

$$H_o: \sigma^2 \leq 0.25 \quad \text{and} \quad H_a: \sigma^2 > 0.25$$

Solution continued . . .

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(41-1)(0.27)}{0.25} = 43.2$$

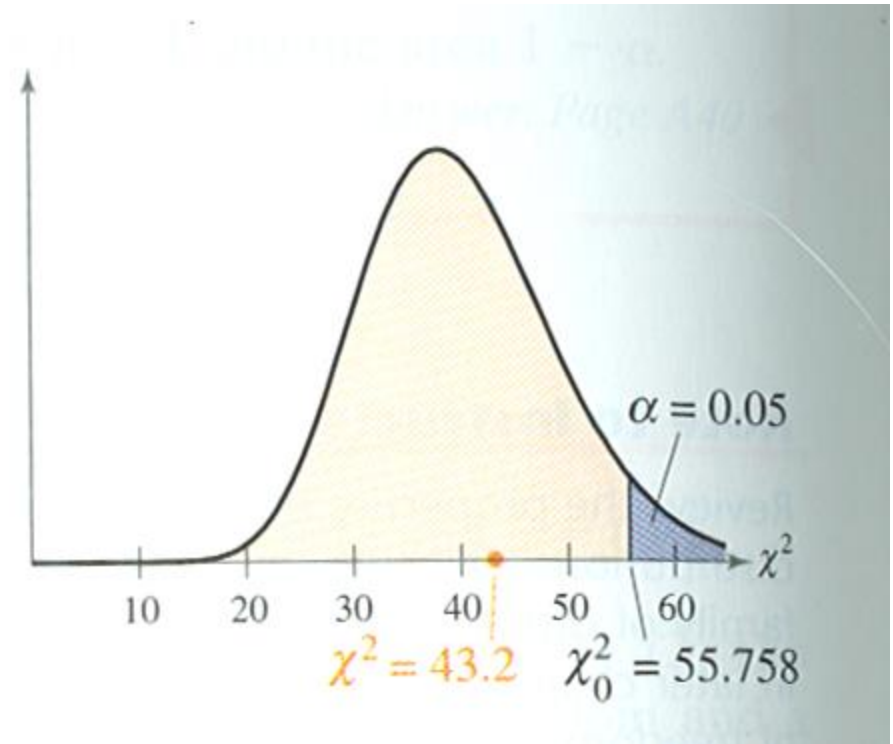
- The graph shows the location of the acceptance region and the standardized test statistic, χ^2 . Because χ^2 is not in the acceptance region, you should decide not to acceptance the alternative hypothesis.

Solution continued . . .

- Because the test is a right-tailed test, the level of significance is $\alpha = 0.05$. There are d.f. = $41 - 1 = 40$ degrees of freedom and the critical value is $\chi^2_{0.05, 40} = 55.758$. The acceptance region is $X^2 > 55.758$. Using the X^2 -test, the standardized test statistic is:

Finding Critical Values for χ^2

- You do NOT have enough evidence to accept that the company is wrong at the 5% significance.



Ex. 2 Using a Hypothesis Test for SD

- A restaurant claims that the standard deviation in the length of serving times is less than 2.9 minutes. A random sample of 23 serving times has a standard deviation of 2.1 minutes. At $\alpha = 0.10$, is there enough evidence to support the restaurant's claim? Assume the population is normally distributed.

Solution

- The claim is “the standard deviation is less than 2.9 minutes.” So the null and alternative hypotheses are :

$H_0: \sigma \geq 2.9 \text{ min}$ and $H_a: \sigma < 2.9 \text{ min}$

Solution continued . . .

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(23-1)(2.1)^2}{2.9^2} \approx 11.54$$

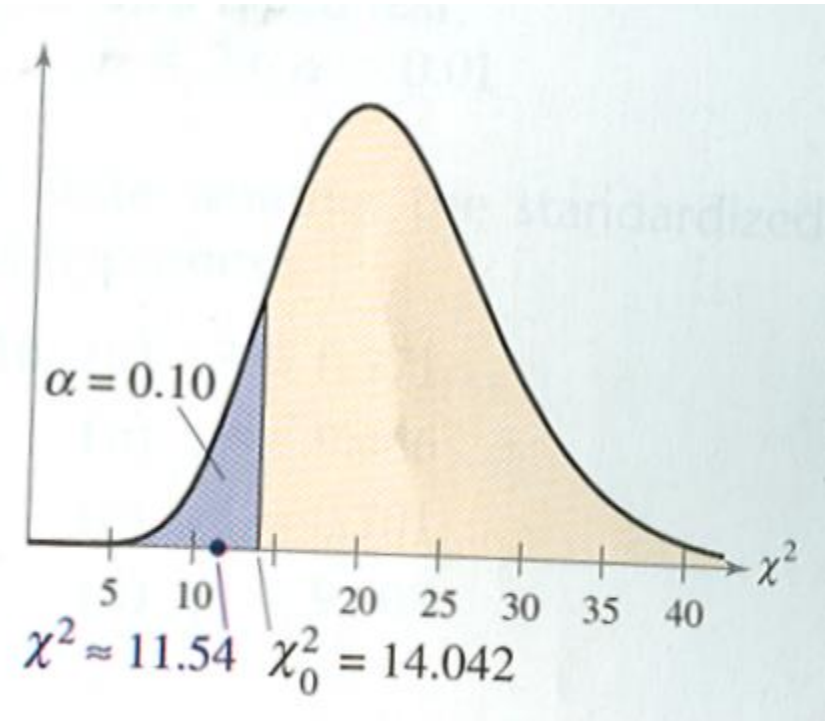
- The graph shows the location of the acceptance region and the standardized test statistic, χ^2 . Because χ^2 is in the acceptance region, you should decide to accept the alternative hypothesis.

Solution continued . . .

- Because the test is a left-tailed test, the level of significance is $\alpha = 0.10$. There are d.f. = $23 - 1 = 22$ degrees of freedom and the critical value is 14.042. The acceptance region is $X^2 < 14.042$. Using the X^2 -test, the standardized test statistic is:

Solution continued . . .

- So, there is enough evidence at the 10% level of significance to support the claim that the standard deviation for the length of serving times is less than 2.9 minutes.



Ex. 3: Using a Hypothesis Test for the Population Variance

- A sporting goods manufacturer claims that the variance of the strength of a certain fishing line is 15.9. A random sample of 15 fishing line spools has a variance of 21.8. At $\alpha = 0.05$, is there enough evidence to say that the manufacturer's claim is wrong? Assume the population is normally distributed.

Solution

- The claim is “the variance is 15.9.” So, the null and alternative hypotheses are:

$$H_0: \sigma^2 = 15.9 \text{ and } H_a: \sigma^2 \neq 15.9 \text{ min}$$

Solution continued . . .

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(15-1)(21.8)}{15.9} \approx 19.19$$

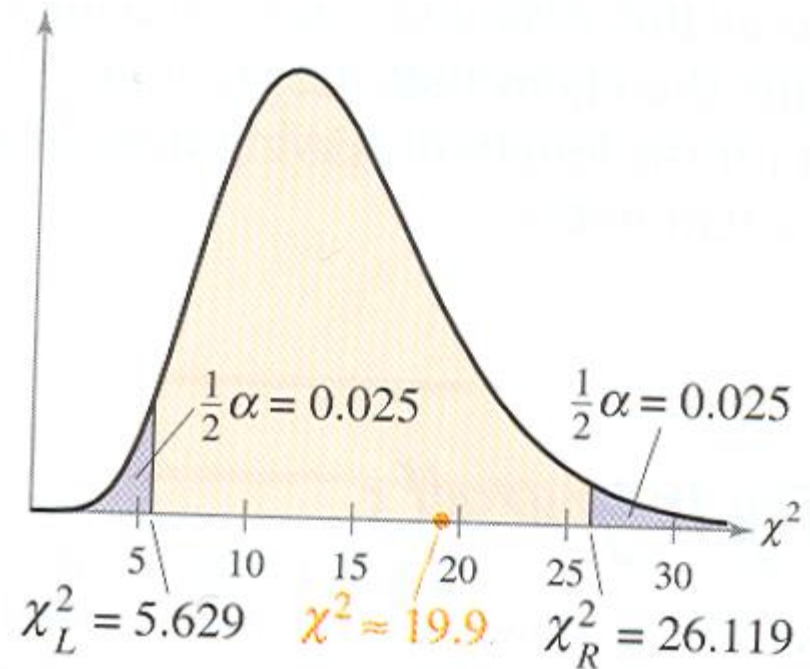
- The graph shows the location of the acceptance regions and the standardized test statistic χ^2 . Because χ^2 is not in the acceptance regions, you should decide not to accept the alternative hypothesis.

Solution continued . . .

- Because the test is a two-tailed test, the level of significance is $\alpha = 0.05$. There are d.f. = $15 - 1 = 14$ degrees of freedom and the critical values are 5.629 and 26.119. The acceptance regions are $\chi^2 < 5.629$ and $\chi^2 > 26.119$. Using the χ^2 -test, the standardized test statistic is:

Solution continued . . .

- At the 5% level of significance, there is not enough evidence to accept the manufacturer's claim- that the variance is 15.9- is wrong.



Study Tip

- Although you are testing a standard deviation, the χ^2 statistic requires variances. Don't forget to square the given standard deviations to calculate these variances.

CONFIDENCE INTERVAL FOR POPULATION VARIANCE σ^2

$$\frac{(n-1)s^2}{\chi_R^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_L^2}$$

right-tail
critical value

left-tail
critical value

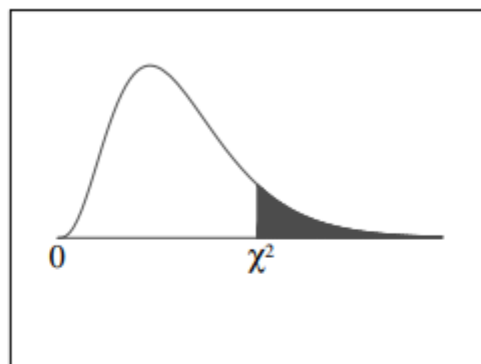
CONFIDENCE INTERVAL FOR POPULATION STANDARD DEVIATION σ

$$\sqrt{\frac{(n-1)s^2}{\chi_R^2}} < \sigma < \sqrt{\frac{(n-1)s^2}{\chi_L^2}}$$

Example

- Candy packs weighing 52 grams are produced, packaged, and sold by a large candy manufacturer. To estimate the standard deviation of all 52-gram packs made by the manufacturer, they randomly selected $n = 10$ packs from the factory line. The random sample yielded a sample variance of 4.2 grams. Use the random sample to derive a 95% confidence interval for σ .

Chi-Square Distribution Table



The shaded area is equal to α for $\chi^2 = \chi^2_{\alpha}$.

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188

Answer

First, we need to determine the two chi-square values with $(n-1) = 9$ degrees of freedom

Now, it's just a matter of substituting in what we know into the formula for the confidence interval for the population variance. Doing so, we get:

$$(9(4.2)/19.023 \leq \sigma^2 \leq 9(4.2)/2.7)$$

Simplifying, we get:

$$(1.987 \leq \sigma^2 \leq 14.0)$$

We can be 95% confident that the variance of the weights of *all* of the packs of candy coming off of the factory line is between 1.987 and 14.0. Taking the square root of the confidence limits, we get the 95% confidence interval for the population standard deviation σ :

$$(1.409 \leq \sigma \leq 3.742)$$

That is, we can be 95% confident that the standard deviation of the weights of *all* of the packs of candy coming off of the factory line is between 1.409 and 3.742 grams.